

Scalable pluvial flood risk assessment: A data-driven framework integrating machine learning (ML) and discrete global grid systems (DGGS H3)

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ARTICLE INFO

Keywords:

Surface-water flooding
Geospatial intelligence
Urban resilience
Discrete global grid system (DGGS)
Random forest
Climate adaptation
Flood hazard mapping

ABSTRACT

The escalating frequency of extreme rainfall events necessitates scalable and dynamic tools for urban flood risk assessment. The primary objective of this study is to demonstrate and evaluate how a machine-learning-derived building-level pluvial flood susceptibility index can be operationalised through multi-resolution aggregation using the H3 Hexagonal Discrete Global Grid System (DGGS). The study builds on the building-level Pluvial Flood Index (PFI_b) developed in the InzureFlood project, which derives susceptibility from historical insurance damage patterns and geospatial predictors.

In the proposed methodology, the analysis domain is partitioned using H3, and building-level scores are aggregated within hexagonal cells to produce a scalable hexagon-level index (PFI_h). Results indicate that this grid-based approach substantially improves computational efficiency, reducing spatial query operations by approximately 98% compared to traditional geometry-based workflows. The resulting maps show spatial patterns that are qualitatively consistent with expected topographic controls, such as depressions and flow paths. However, quantitative analysis of resolution effects reveals a critical trade-off: while overall susceptibility levels are preserved, aggregation increasingly smooths local extremes. A Jaccard Similarity Index of 0.14 between street-level (R13) and neighborhood-level (R10) hotspots indicates that most fine-scale susceptibility hotspots are not visible at intermediate scales.

These findings imply that coarser resolutions may support regional strategic screening, whereas high-resolution grids are necessary for safety-critical tasks such as emergency response and local mitigation. The framework provides a scalable and updateable representation that supports consistent multi-scale communication from micro-scale screening to macro-scale planning.

1. Introduction

Urban flooding predominantly occurs during intense rainfall events in densely populated areas with inadequate drainage and sewer systems [1,2]. This phenomenon, termed *pluvial flooding*, differs from *fluvial flooding*, which arises from overflowing rivers. Although

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pluvial floods often cause less damage per event than fluvial floods, their higher frequency can result in substantial cumulative losses [3,4]. Recent work on urban pluvial flood susceptibility mapping also shows how machine-learning models can identify flood-prone locations while supporting more targeted mitigation planning [5].

These social and economic costs are likely to escalate over time, as urban populations grow and climate change increases the probability and severity of extreme rainfall events [6]. A combination of climatic and socioeconomic changes is expected to increase flood risk in urban areas in the future [7,8]. Traditionally, infrastructure-based drainage solutions have been criticized by many researchers as they lack sufficient flexibility, which is needed to adapt to critical events which are expected in the future [9,10]. Thus, the established design practice must be revised, which includes more use of non-piped, nature-based solutions (NBS) to meet the impact that more extreme weather conditions have on urban areas [8].

However, to ensure that urban densification and expansion do not exacerbate flood risks, systematic and sustainable spatial planning is essential. Regulatory frameworks that guide neighborhood design and infrastructure standards can further reduce flood vulnerability [11,12]. In already built-up districts, public spaces such as parks, pedestrian zones, and bicycle paths can function as open runoff systems, although denser urban areas may still require a blend of open and closed drainage solutions. A critical challenge in this context is the spatiotemporal variability of flood risk, driven by change in precipitation distribution and land-use dynamics. Enhanced awareness of how urban expansion interacts with emerging climate risks is pivotal for decision-makers involved in risk and vulnerability assessments [13].

Flood-prone areas can be quantified using susceptibility indices derived from historical observations or model outputs [14,15]. Such indices enable comparative assessment across locations and support the identification of priority areas for mitigation and preparedness. Because flood-risk information is used differently across stakeholder groups and decision contexts, susceptibility outputs should be communicated at multiple spatial scales, from building/street-level information for property-related decisions to neighborhood and city scales for planning and preparedness [14,16].

Disaster risk reduction (DRR) depends on actionable and accessible spatial information that supports preparedness, mitigation prioritisation, and response planning. Susceptibility maps can be operationalised to delineate comparatively safer and less safe areas and to support evacuation-oriented decision support (e.g., Karpouza et al. [17]). Building on this need, this study proposes a scalable framework for communicating building-level pluvial flood susceptibility through multi-resolution hexagonal maps using the H3 DGGS [18]. The partitioning of space into discrete units is a core principle of spatial analysis, and DGGS frameworks provide consistent, hierarchical spatial references that enable aggregation and interpretation across multiple scales [19]. In particular, Uber's H3 spatial indexing system offers uniform spatial coverage, minimizes edge effects, and supports efficient neighborhood queries and cross-scale aggregation [17,18]. Originally developed for fleet optimization, H3 is now adopted across diverse application areas, including disaster management [20], and it can also support anonymization of sensitive geospatial information by reporting susceptibility at cell level rather than at exact building locations, which is increasingly relevant under privacy regulations such as the European Union's General Data Protection Regulation (GDPR).

H3 has previously been recognized as tool for hydrological analyses, flow direction, watershed delineation, multi-resolution terrain modeling [21,22]. Other environmental applications, such as wildfire distribution and emergency-response routing, reinforce their effectiveness in preserving spatial continuity and enabling efficient adjacent operations. Recent literature emphasizes the urgent need to shift from reactive disaster management to proactive emergency planning strategies. Effective disaster risk reduction (DRR) requires tools that not only identify risks but support the strategic allocation of resources before an event occurs [23].

Random Forest and related ensemble methods have demonstrated strong performance in urban pluvial flood susceptibility mapping when combining terrain-based predictors with built-environment and drainage-network variables [15,24]. This study builds on established modelling logic by using a building-level susceptibility index (PFI_b) derived from integrated geospatial predictors and communicating it through multi-resolution H3 aggregation.

This article investigates the feasibility of communicating building-level pluvial flood susceptibility using a scalable, multi-resolution mapping approach based on the H3 discrete global grid system. The study builds on an existing machine-learning-derived building index (PFI_b) [25,26] and examines how these values can be aggregated within H3 hexagonal cells to produce a consistent hexagon-level representation (PFI_h) across spatial scales. Both PFI_b and PFI_h are treated as relative susceptibility indicators intended for screening, comparison, and prioritisation of areas that may warrant closer assessment, rather than as probabilistic estimates of flood occurrence or expected loss. The contribution of the paper is to demonstrate how H3-based aggregation enables interpretable and operational flood susceptibility mapping that can support urban planning and disaster risk reduction across decision-relevant scales.

In this paper, **hydrology** refers to the transformation of precipitation (and snowmelt) into runoff volumes, whereas **hydraulics** refers to the routing of water through the surface and drainage network, producing water depths and flow velocities. The susceptibility framework presented here targets likely impact locations (buildings) based on observed damage patterns and geospatial predictors, rather than event-specific inundation depths. It should therefore be interpreted as a decision-support layer for screening and prioritisation, distinct from hydrologic/hydraulic simulation models used for event-based forecasting and design analysis.

The aim of this paper is to develop and demonstrate a scalable, operational method for transforming PFI_b into multi-resolution H3 based indices (PFI_h), and to evaluate how well this representation supports decision-making across spatial scales relevant to urban flood risk management. Inspired by disaster risk mapping studies using H3 to reveal intra-urban patterns [27], the following research questions are posed:

Can building-level scores (PFI_b) produced by the machine-learning model be aggregated within H3 cells to yield a scalable hexagon-level index (PFI_h)?

Accordingly, this study addresses three subsidiary research questions.

- *Spatial Suitability*: Do H3 accurately reflect spatial clusters of flood susceptibility at different scales?
- *Analytical Usability*: Does aggregating PFI_b into PFI_h preserve key risk information and aid interpretation for diverse stakeholders (e.g. insurers, planners, emergency managers)?
- *Operational Feasibility*: Can the resulting H3 based visualizations support dynamic updates and decision-making, allowing real-time recalibration as new data (e.g., flood claims or land-use changes) arrive?

2. Materials and methods

This study builds on the InzureFlood dataset and the building-level Pluvial Flood Index (PFI_b) framework described in Svellingen and Torgersen [25,26] (2023, 2024). This study summarise only the elements required to interpret PFI_b in the present context and focus on the methodological contribution of aggregating PFI_b into a multi-resolution hexagon index (PFI_h) using the H3 discrete global grid system. Accordingly, this section describes the data basis, summarises the relevant PFI_b components, and details the H3-based aggregation workflow.

2.1. Pluvial Flood Index for individual buildings (PFI_b)

The building-level pluvial flood susceptibility index (PFI_b) used in this study was originally developed within the InzureFlood pilot project by 7Analytics AS in 2024 in collaboration with Fremtind, one of Norway's largest insurance companies. InzureFlood established the underlying machine-learning workflow and feature set for estimating susceptibility from observed historical damage patterns and geospatial predictors, and demonstrated its operational relevance for screening and prioritisation in Norwegian contexts [25]. The present study does not redevelop or retrain the underlying building-level model; rather, it builds on the existing PFI_b output and focuses on how this information can be communicated and operationalised across spatial scales using the H3 discrete global grid system.

Random Forest was selected as the base model due to its robustness to non-linear relationships, mixed predictor types, and high-dimensional geospatial feature spaces, which makes it well-suited for urban flood susceptibility mapping [15]. Previous studies have shown that model performance improves when traditional terrain and geomorphological predictors are complemented with drainage-network descriptors such as sewer density and proximity to storm drains. For example, a Seoul case study reported strong discrimination performance using Random Forest when integrating drainage-related variables with geomorphological features [28]. Following this established approach, the predictor set used for PFI_b combines upstream catchment descriptors (e.g., slope, curvature, and flow-path length) with built-environment variables (e.g., building type, roof age, and impervious surface coverage). This integrated feature design enables the model to represent both hydrologic controls on runoff concentration and infrastructure-related controls on local drainage capacity, which are critical for pluvial flood susceptibility in dense urban environments.

PFI_b is generated by a machine-learning model designed to identify patterns across variables that describe each building's location and its surrounding environment. The model uses more than 250 quantifiable predictors capturing terrain, land cover, drainage-related characteristics, and other relevant geospatial attributes. During the InzureFlood project, PFI_b was constructed to represent relative building vulnerability by learning from historical damage records in combination with these spatial predictors [25]. While the original implementation was developed for Norway, the general modelling concept is transferable in principle to other settings where comparable input datasets exist, although robust application elsewhere requires appropriate data availability and, where necessary, local calibration.

Insurance claim data in Norway provide building-level records of flood-related damages, including addresses, event dates, and associated costs. These data have been standardized since 2006 and are administered by Finance Norway [29]. Within the InzureFlood project, such claims were used as the primary empirical basis for training the PFI_b model to distinguish buildings with documented flood damage from those without recorded loss. As implemented, PFI_b therefore represents susceptibility inferred from historical conditions and observed damage patterns, and it is not explicitly conditioned on rainfall return periods or projected future climate scenarios. Extending the approach toward future-oriented assessments would require coupling the framework with scenario-based precipitation datasets and climate projections (e.g., IPCC [6]) to evaluate how changing rainfall extremes may influence pluvial flood susceptibility.

The PFI_b model was trained on Norwegian insurance-claim data and corresponding geospatial predictor variables within the InzureFlood project [25]. In the present study, the trained model is applied to buildings in the selected New York case area without local retraining, by computing the same predictor variable set using locally available datasets. The resulting building-level PFI_b scores are then used as input for aggregation within the H3 grid system to produce the hexagon-level index (PFI_h) across multiple resolutions. This enables evaluation of how multi-resolution representation affects interpretability and hotspot persistence, while keeping the underlying building-level susceptibility model fixed.

As the model is transferred without local retraining, PFI_b in the New York case area should be interpreted as a relative susceptibility proxy based on the learned relationships, and performance may be affected by differences in exposure characteristics, reporting practices, and data generating processes between regions.

The methodological focus of this study is the representation and aggregation of building-level indices within the H3 DGGS. To illustrate the multi-resolution mapping and aggregation workflow, we present example maps for a selected urban case area. These examples are intended to demonstrate the behavior of the H3 based aggregation and the interpretability of PFI_h across resolutions, rather than to provide an independent predictive validation for a new geographic setting.

This methodology implies a critical conceptual insight: The catchment is not a static input to the model, but rather a derived feature

that is itself a function of the building location and local topography. In other words, each PFI_b implicitly encodes a tailored hydrological unit surrounding the structure, generated through geospatial analysis. It is from this hydrological unit that most features are extracted. Hence, rather than using fixed catchment boundaries like river basins or administrative areas, the ML approach dynamically constructs the hydrological footprint most relevant to each building.

The feature extraction logic follows three tiers (Fig. 1).

- *Local parameters* (within a 10–30m buffer): building footprint, construction year, material types, impervious surface coverage, proximity to roads and culverts.
- *Upstream catchment parameters*: mean elevation, slope variance, curvature, drainage density, dominant land-use categories, upstream flow path length.
- *Contextual parameters*: distance to nearest river, flow-path length, sinks (accumulation area), floodplain proximity, urbanization index of the surrounding neighborhood.

All predictor variables, including upstream catchment statistics, are computed **per building** by delineating a local contributing area based on the building location and the surrounding topography. The resulting metrics are stored as continuous numeric features (e.g., mean slope, curvature, flow-path length) used directly by the Random Forest model.

These inputs are synthesized into a feature vector and fed into a Random Forest model, an ML algorithm known for its robustness in handling high-dimensional, non-linear datasets [30,31]. The algorithm builds an ensemble of decision trees using bootstrap sampling and majority voting, iteratively refining its performance through out-of-bag validation [24].

Initially, the random forest model in InzureFlood was trained on approximately 1.3 million buildings across Norway, encompassing both flooded and non-flooded structures. This binary classification framework (flooded vs. non-flooded) allowed the model to discern complex patterns indicative of pluvial flood risk. Once trained, the model produces a PFI_b value (ranging from 0 to 1) for each building, reflecting the severity of flooding under conditions like those recorded historically.

2.2. H3 – Hexagonal Discrete Global Grid System (DGGS)

To support efficient multi-resolution spatial processing, this study uses Uber's H3 DGGS. Hexagonal grids provide a uniform neighborhood structure and consistent center-to-center distances, which reduces directional artifacts in local spatial analysis compared with square tiling [32,33]. In the workflow, building locations are represented by H3 cell identifiers, enabling spatial joins and neighborhood queries to be performed directly on discrete cell IDs rather than through computationally expensive geometry intersections [34,35]. The hierarchical structure of H3 further enables consistent analysis and visualization across multiple resolutions within the same spatial reference system [36–38]. Representing outputs at hexagon-cell level instead of exact building coordinates also reduces spatial precision in shared products, supporting privacy-aware handling of geospatial risk information. H3 aggregation reduces precision, but does not guarantee anonymization if auxiliary datasets enable re-identification; appropriate resolution choice remains necessary.

Building locations (latitude/longitude) are converted to H3 cell identifiers using the H3 point-to-cell indexing function. H3

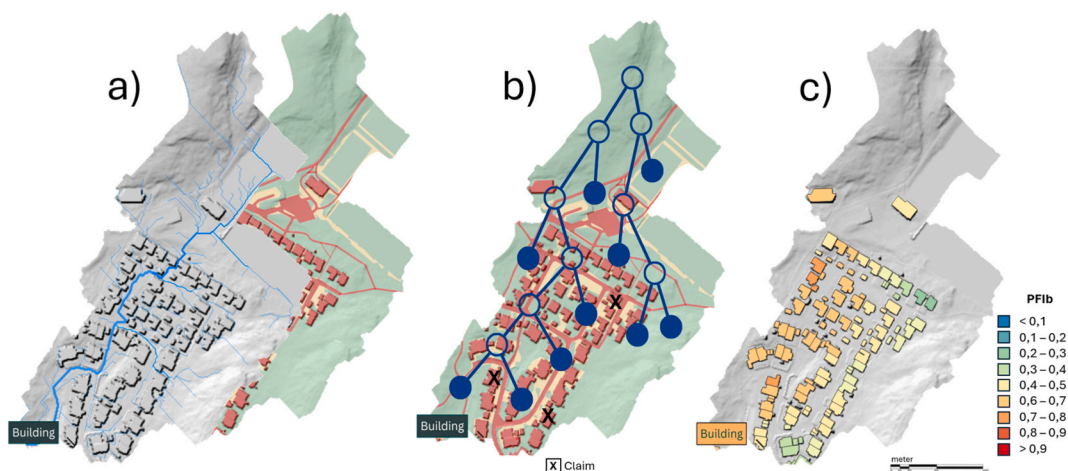


Fig. 1. The highlighted area indicates the broader catchment context used to compute local terrain and upstream predictor variables for the “Building”. Grey shading represents the underlying digital elevation model (DEM) used to describe terrain and flow routing, while the colored background represents land-use/land-cover classes. (a) The PFI_b is generated by extracting geospatial parameters (e.g. elevation, land cover, proximity to drainage) for each building. (b) These features, combined with historical flood claims data (x), are preprocessed and used to train a Random Forest model (illustrated by blue lines and circles), which learns the relationship between building characteristics and flood susceptibility. (c) The output of the model is the PFI_b , representing the relative flood risk for each building.

provides a hierarchical grid with 16 global resolution levels, where each parent cell is subdivided into seven child cells at the next finer resolution, enabling consistent multi-scale representation and cross-scale aggregation [32,33]. The selected H3 resolutions are illustrated in Fig. 2 to support comparison of susceptibility patterns across spatial scales.

The H3 hexagonal grid system, conceptually shown in Fig. 2a, comprises 16 resolution levels globally [39]. Each resolution subdivides a parent cell into seven child cells, enabling consistent multi-scale analysis and visualization. Fig. 2b highlights the average area corresponding to each level (calculated with a spherical Earth model using the WGS84/EPSSG:4326 authalic radius).

H3 divides the globe into hexagonal cells at different resolutions [40], commonly referred to as resolution (Fig. 2b). Higher numbers (resolution R11-R15) offer hexagonal cells for *micro* level, suitable for the most detailed, localized assessments [21]. The *macro* level (resolution R0-3) covers hexagonal areas, suitable for broad-scale analyses (e.g., national or regional planning).

According to Zolghadr-Asli et al. [41], data privacy and the ethical collection of data are both critical concerns. This data should be collected and shared responsibly. Protecting the rights of individuals and organisations requires clear guidelines and regulatory oversight. Particularly, building-level information, such as exact addresses or coordinates, can raise privacy risks if linked back to individuals. Representing susceptibility outputs at the hexagon-cell level rather than at exact building coordinates also reduces spatial precision in shared products, thereby supporting privacy-aware dissemination of geospatial risk information.

2.3. Aggregation of building-level PFI_b to hexagon-level PFI_h using H3

In this study, PFI_b values are aggregated and adapted to produce multi-scale PFI_h maps. While the random forest model outputs PFI_b in the range [0, 1], translating these results into practical decision-support tools often requires aggregation at broader spatial scales. The H3 grid system addresses this need by partitioning a given region into discrete hexagonal cells. For each hexagon, the PFI_h is calculated by averaging the PFI_b values of the buildings falling within that cell (Equation 1):

$$PFI_h = \frac{\sum_{i=1}^n PFI_{b,i}}{n}, PFI_h \in [0, 1]$$

where $PFI_{b,i}$ is the building-level index for the i -th building, and n is the number of buildings in the hexagon. Because mean aggregation can assign similar PFI_h values to cells with very different numbers of buildings, the number of buildings per cell (n) is reported alongside PFI_h to support interpretation. The default aggregation uses the mean of building-level PFI_b values within each H3 cell, producing an index in the range 0–1 that can be mapped and used for relative ranking of susceptibility zones. To support applications where localized extremes are important, complementary within-cell summaries may also be calculated, including $sum(PFI_b)$, the share of buildings above a high-susceptibility threshold, the 95th percentile (P95), or the maximum. As a practical decision rule, mean-based PFI_h is most appropriate for strategic screening and stable area-wide overviews, whereas extreme-preserving summaries (P95, maximum, or share-above-threshold) are better suited for hotspot-sensitive and safety-critical use cases where high local susceptibility

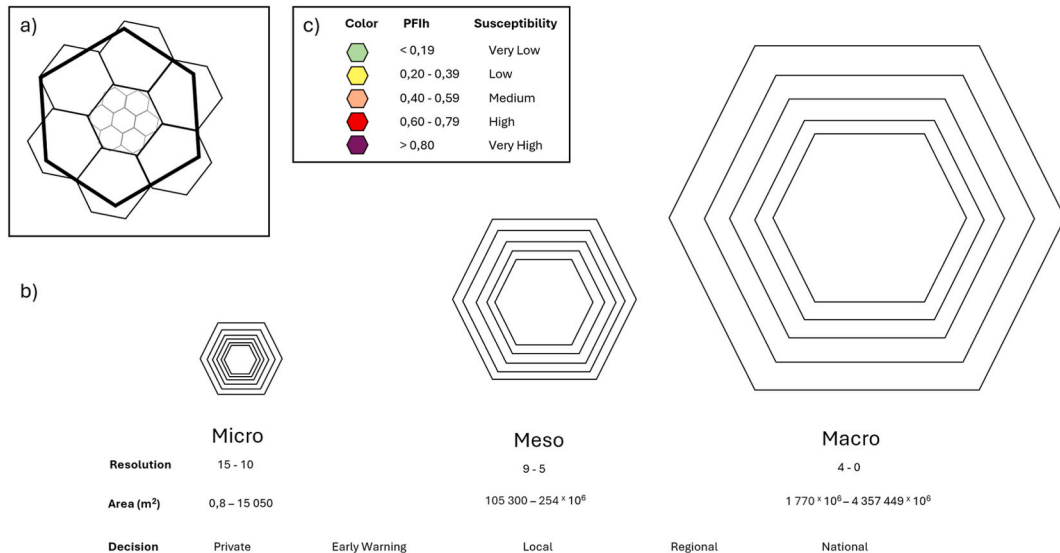


Fig. 2. a) Illustration of the relationship between a parent hexagonal cell and its seven child cells, emphasizing the recursive and nested structure of the H3 system. b) Representation of H3 grid resolutions across three spatial scales, Micro, Meso, and Macro, each associated with distinct decision-making contexts. The hierarchical structure allows seamless transitions from fine-grained analyses (e.g., building-level or neighborhood-scale) to coarse regional or national assessments. c) Classification scheme for relative pluvial flood vulnerability (PFI_h), based on five discrete color intervals, ranging from Very Low (<0.19) to Very High (>0.80). This scale supports rapid risk interpretation and consistent communication across stakeholder groups.

should not be smoothed by aggregation [42–44].

By aggregating PFI_b within each H3 cell, we generate a dynamic pluvial flood susceptibility map that can be queried and visualized across multiple resolutions, supporting applications from strategic screening to local preparedness. This map can be displayed and analyzed at different zoom levels, thereby informing a diverse range of decision-making contexts, from regional spatial planning to localized emergency preparedness [45].

PFI_h values are visualized using a five-class palette representing Very Low, Low, Medium, High, and Very High pluvial flood vulnerability (Fig. 2c and 3b). The class labels follow the terminology recommended by the Norwegian Directorate for Civil Protection (DSB) [46], supporting consistent interpretation across stakeholder groups and map scales. The palette is designed to be intuitive, with progressively stronger visual intensity indicating increasing susceptibility, thereby enabling rapid identification of areas of concern and supporting communication across both technical and non-technical audiences.

The threshold calibration is not the primary objective of this study. Instead, the class cut points are intended as indicative and are informed by empirical comparisons reported in Svellingen and Torgersen [26], where locations with repeated historical damages were associated with higher index values, suggesting a practical separation around $PFI_h \approx 0.6$. The resulting map classes should therefore be interpreted as relative screening and prioritisation categories, rather than as calibrated probability-of-damage estimates.

3. Results

The H3 aggregation produces a coherent susceptibility surface across resolutions, enabling comparison between city-scale patterns and fine-scale hotspots within the same indexing framework (Fig. 4). Building-level PFI_b values were first generated by applying the trained model and then aggregated to PFI_h within H3 cells to evaluate scale effects and mapping behaviour across resolutions. The New York dataset serves as a testbed for the *computational scalability* and *visual aggregation* capabilities of the H3 framework. While the susceptibility patterns align with topographic expectations, localized retraining is a prerequisite for operational insurance underwriting in this specific region.

3.1. Spatial patterns and topographic alignment

The New York case study presents a broad-scale municipal overview (Fig. 4); Fig. 4a narrows in on the case area center and adjacent areas, and Fig. 4b and c zooms further to an urban district. Notably, the susceptibility maps reveal how susceptibility hotspots may be distributed within each jurisdiction. Zones classified as Very High vulnerability frequently coincide with terrain configurations that are plausibly conducive to pluvial water accumulation. In this study, local depressions and flow-path constraints are not mapped explicitly as standalone geomorphic objects; instead, they are represented indirectly through building-specific terrain predictors used by the ML model (e.g., curvature, local slope variability, and flow-path length and sinks (accumulation area)), which quantify convergent topography and potential drainage limitations in the surrounding feature-extraction area. By contrast, lower-indexed zones tend to occur on comparatively elevated, flatter, or better-drained surfaces. Overall, the observed spatial patterns are consistent with the expected influence of topography on pluvial flood susceptibility.

Fig. 4 also illustrates that high PFI_h values tend to cluster in areas near flowing or collected surface water, such as depressions or narrow drainage pathways. The areas classified as high susceptibility show spatial patterns that are qualitatively consistent with expected pluvial flood controls, including local depressions, convergent terrain, and constrained overland flow paths. Visual inspection indicates that the mapped susceptibility patterns align with topographic configurations that are plausibly conducive to pluvial water accumulation [26].

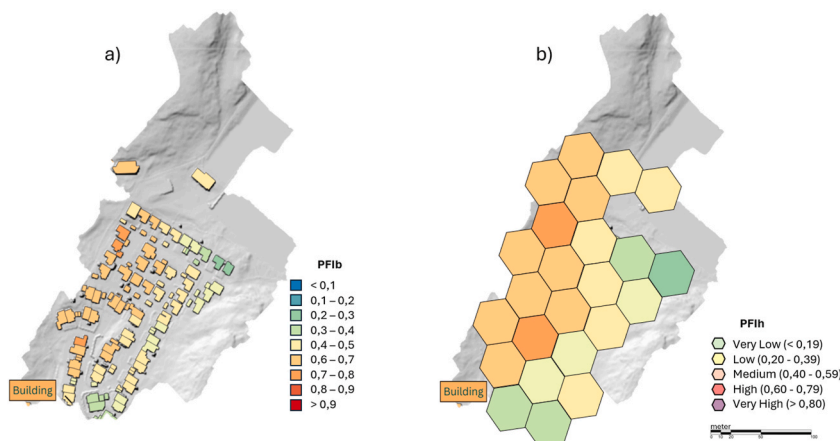


Fig. 3. The Pluvial Flood Index PFI_b to PFI_h . Transition from (a) building-level risk (PFI_b) to (b) hexagonal aggregation (PFI_h).

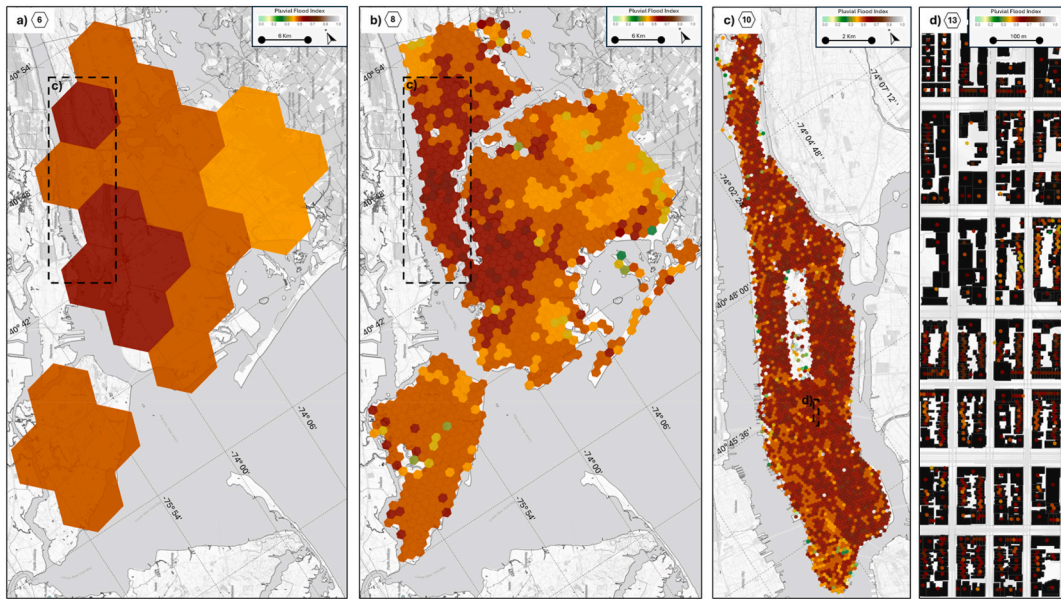


Fig. 4. Multi-resolution pluvial flood susceptibility mapping using H3 for the New York City administrative boundary (NYC DCP). The hexagon-level index PFI_h is computed by aggregating building-level PFI_b values within each H3 cell and is visualized using a consistent color palette across resolutions. (a) H3 resolution R6, providing a city/borough-scale overview for regional screening and cross-borough comparison; the location of zoom area C is indicated. (b) H3 resolution R8, providing neighborhood-scale detail suitable for identifying spatial clusters and supporting local mitigation and drainage-upgrade targeting; zoom area C is indicated. (c) H3 resolution R10, providing a district/corridor-scale view; zoom area D indicates the location of the illustrative inset shown in panel (d). (d) H3 resolution R13, showing a conceptual parcel/street-scale inset (illustrative zoom) to demonstrate the level of spatial detail achievable at high resolutions and the emergence of sub-block susceptibility patterns. Across panels, the same underlying aggregation procedure enables seamless interpretation from city-wide patterns to street-level susceptibility hotspots within a unified spatial indexing framework.

3.2. Quantitative analysis of resolution effects

To evaluate the stability of the PFI_h across scales, we performed a quantitative analysis on the New York case study area, examining the same resolutions visualized in Fig. 4 (H3 resolution R6, R8, R10, and R13). This analysis assessed both the statistical distribution of index values and the spatial persistence of susceptibility hotspots as the H3 resolution coarsens.

As H3 resolution coarsens, the aggregated index exhibits a pronounced smoothing effect. At finer resolutions (R12–R13), building- and street-level contrasts remain visible, and the distribution retains pronounced "shoulders" and tails, indicating that a substantial number of hexagons capture the very-low and very-high PFI_h extremes essential for micro-scale assessment.

However, as the grid coarsens, within-cell heterogeneity is averaged out. As shown in Fig. 5a, the variance decreases typically with resolution. At the finest resolution applied to New York (R13, $n \approx 1.2$ million buildings (hexagons)), the distribution spans the full susceptibility range (0.0–0.8). By the municipal scale (R6, $n = 41$ hexagons), the distribution compresses significantly into a narrow band (0.45–0.55), effectively filtering out the localized extremes.

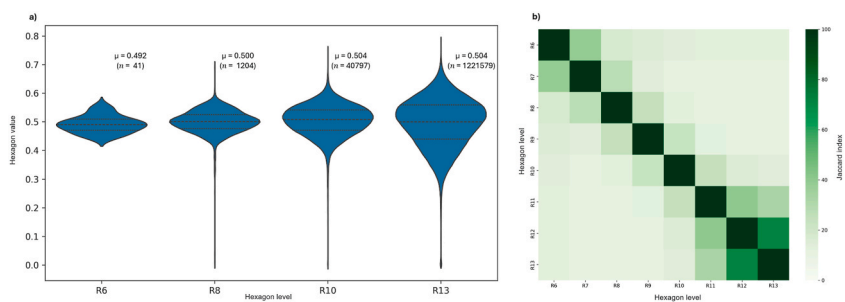


Fig. 5. Quantitative analysis of resolution effects. (a) Frequency distribution of PFI_h across H3 resolutions (R6–R13). Violin plots showing the density of PFI_h values. While the mean susceptibility ($\mu \sim 0.50$) remains stable, the variance decreases sharply with coarsening, illustrating the loss of extreme values at R6 compared to the full range visible at R13. (b) Hotspot Persistence. Heatmap of the Jaccard Similarity Index (%), quantifying the spatial overlap of high-susceptibility hotspots (top 10%; sensitivity to alternative cutoffs discussed in text) between resolutions. The low overlap between R13 and R10 ($J = 0.14$) demonstrates that fine-scale susceptibility hotspots are significantly diluted when aggregated to coarser grids.

Notably, despite this compression, the central tendency remains robust. The mean susceptibility score (μ) remains consistent across all resolutions, fluctuating only slightly between 0.492 at R6 and 0.504 at R10. This confirms that the total "system risk" is conserved during aggregation, even as local details are obscured.

This statistical attenuation has direct spatial consequences. To quantify whether high-susceptibility areas identified at fine scales remain visible after aggregation, the Jaccard Similarity Index (J) was calculated for the top 10% of susceptibility zones across resolutions (Fig. 5b). A sensitivity check using alternative hotspot cutoffs (top 5–20%) produced the same qualitative pattern of low hotspot overlap between fine and intermediate resolutions (results not shown).

Hotspot sets were compared across resolutions by mapping each fine-resolution H3 cell to its corresponding parent cell at the coarser resolution using the H3 hierarchical indexing relationship. The resulting heatmap reveals a rapid spatial decay as the difference in resolution increases. While adjacent resolutions (e.g., R12 vs. R13) exhibit strong agreement, the overlap between the street-level R13 and the neighborhood-level R10 drops to $J = 0.14$ (14%). This indicates that 86% of the specific high-susceptibility hotspots identified at R13 are diluted or "lost" when visualized at the R10 neighborhood scale. This lack of persistence underscores the operational risk of relying solely on intermediate resolutions for safety-critical tasks, as they may generate false negatives by averaging small, severe risk clusters with surrounding safe terrain.

4. Discussion

4.1. Verification of high-risk zones

Independent claim-level flood damage data were not available for the New York demonstration area. Therefore, the evaluation presented here focuses on (i) computational scalability and (ii) qualitative consistency with topographic expectations. Quantitative validation against historical insurance claims has previously been demonstrated for Norwegian municipalities in Svellingen and Torgersen [26]. Where independent insurance-claim data were available, higher index values aligned with a higher frequency of recorded flood damages. In the Fredrikstad validation study, the mean catchment zone index increased from <0.55 in areas with zero recorded damages to >0.60 in areas with three or more recorded damages, with the same positive trend observed across multiple validation datasets [26]. Although some municipalities may lack comprehensive flood damage records, especially in regions with minimal reported incidents, aggregating PFI_b values helps mitigate data sparsity. Comparing adjacent areas with varying claim histories allows for cross-validation using proxy indicators like topography, land cover, and expert knowledge [47].

Demonstrating agreement between historically documented flood impacts and elevated PFI_h values can provide a transparent credibility check for susceptibility mapping and may support trust and practical uptake in risk communication. Such alignment has been demonstrated in Norwegian applications reported in Svellingen and Torgersen [26]. In the present study, the New York case area is instead used primarily to demonstrate the H3-based multi-resolution aggregation workflow and to examine how spatial resolution influences the visibility and persistence of susceptibility patterns across scales.

In this study, the color palette remains consistent across different zoom levels, drawing upon intervals derived from historical damage records [26]. Maintaining the same color scheme regardless of scale promotes interpretive coherence and aids stakeholders in quickly recognizing zones with Very Low, Low, Medium, High, and Very High vulnerability to pluvial events. However, future work could explore alternative weighting strategies, such as factoring in building area or establishing threshold values, to refine the aggregated PFI_h score for specific applications, e.g., property-level risk underwriting or neighborhood-level drainage planning.

4.2. Using H3 hexagons as unit to aggregate flexible susceptibility maps

This study introduces a method for aggregating PFI_b into PFI_h to construct scalable and visually coherent susceptibility maps. The hexagonal structure minimizes edge effects inherent in square or rectangular grids, as hexagons provide uniform center-to-center distances and rotational symmetry [21,32]. Beyond geometry, H3 provides a hierarchical spatial index with consistent neighborhood structure, efficient k-ring (k steps (edges) of a given hexagon) adjacency operations, and seamless aggregation/disaggregation across resolutions, which simplifies scalable computation and multi-scale communication compared with ad-hoc square tilings.

The H3 framework allows seamless scaling from neighborhood-level (fine resolution) to municipal or regional resolution (coarser resolution) (Fig. 2), enabling users to select an appropriate level of granularity for their specific decision-making context [18]. The resolution level determines the area covered by each hexagon, which in turn influences the PFI_h calculation. In practice, finer resolutions yield more precise representations of localized flood risks but may be susceptible to data noise, misreporting, and potential privacy concerns. Conversely, coarser resolutions (larger hexagons) aggregate risk values over wider areas, potentially obscuring smaller pockets of heightened vulnerability. Thus, identifying an optimal zone size depends on the decision-making context, data availability, and the desired balance between granularity and generality.

Moreover, H3 cells can serve as an integrative spatial reference for overlaying susceptibility with complementary layers such as socioeconomic indicators, critical infrastructure, and demographic information, supporting multi-criteria prioritisation in urban planning [37,38]. While a single DGGS reference can simplify such integration, practical adoption involves trade-offs. Many environmental datasets are distributed as rasters on square grids, and translating these products to an H3/DGGS representation requires preprocessing (e.g., raster-to-cell quantization or resampling), which adds computation and can introduce resampling artifacts [38]. H3 is therefore most realistic as an interoperability layer for aggregation, indexing, and querying across heterogeneous datasets, rather than as a full replacement for established raster/grid storage formats [37,38].

Zones at risk for pluvial floods can be rendered across multiple spatial scales using the grid-based system (Fig. 3). These

visualizations highlight potentially flood-prone areas and thereby offer stakeholders, from urban planners and municipal engineers to insurers, a systematic way to identify and prioritize intervention sites.

Additionally, examining how PFI_h values correlate with observed damages or field surveys at various resolutions helps researchers assess whether the random forest model maintains consistent predictive power across spatial scales. Such multi-level verification can highlight scaling effects, where the model might excel at local predictions but exhibit diminished performance at macro scales, or vice versa [48]. This type of analysis supports targeted retraining or data-collection initiatives. For instance, if specific hexagons consistently exhibit high model uncertainty, further surveys or refined topographical studies may be warranted before finalizing risk mitigation measures in these areas.

Because PFI_b already includes catchment-level characteristics specific to each building, general catchment delineation becomes less necessary. Instead, using hexagons ensures that zones of vulnerability are displayed where the risk exists, avoiding the over-generalization often seen in large catchment boundaries. Traditional hydrological planning frequently communicates flood risk using aggregated catchment-based units, which can mask fine-scale spatial heterogeneity within dense urban environments. Event-based rain-on-grid and urban inundation models address a different need by simulating specific storms and return-period scenarios to estimate flood dynamics [49]. The present study instead focuses on data-driven building susceptibility, using machine learning trained on observed damage patterns and geospatial predictors to identify vulnerable buildings and communicate this information consistently across map scales.

4.3. Contrasting hexagons with watershed-based approaches

While watershed boundaries are foundational in hydrological modelling, pluvial flood impacts in dense urban environments are often governed by microtopography, impervious surfaces, and engineered drainage pathways, producing strong spatial contrasts within a single catchment. Hydrologic and hydraulic approaches can represent this complexity in considerable detail, including preferential urban flow routes and flood pathways [50]. However, risk information is frequently communicated using coarse catchment-unit summaries, which may mask intra-urban heterogeneity; a hexagon-based representation offers a practical way to communicate localized contrasts consistently across multiple map scales. This challenge is also reported in operational urban-flood risk tools: in an Oslo case, aggregating results by main catchments produced a misleading prioritisation because catchment sizes varied substantially, and the authors therefore recommended using smaller, more equally sized subareas for communicating risk patterns [51]. The proposed framework is not intended to replace hydrologic rainfall-runoff modelling or hydraulic inundation simulation.

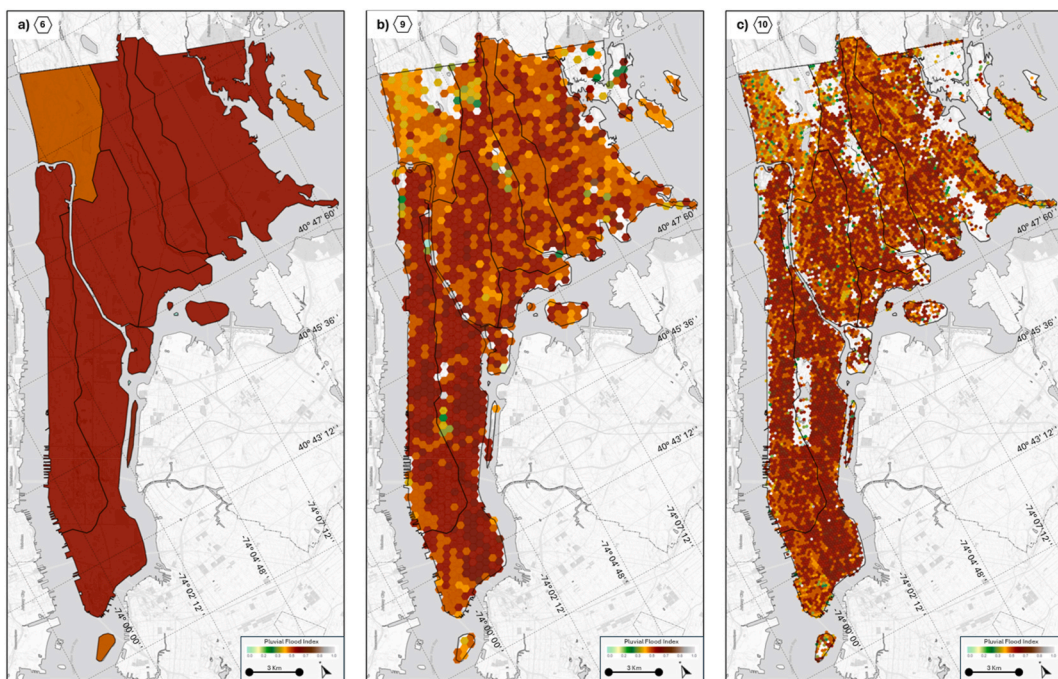


Fig. 6. Catchment vs. H3 representations of pluvial-flood vulnerability (same colour ramp across panels) from New York area. a) USGS HUC-12 subwatersheds: hydrologically consistent polygons that preserve flow context but, at this grain, average out intra-urban variation and mask localized susceptibility hotspots (cells with high PFI_b values indicating high relative propensity for impacts). b) H3 resolution R9 over the same domain: uniform hexagons reveal neighborhood-scale contrasts. c) H3 resolution R11: finer hexes expose street- and block-level susceptibility hotspots that are hidden in catchment maps. Values in (b–c) are the same building-level PFI_b aggregated to hexagons (PFI_h), enabling fair cross-scale comparison. Note: Open/white areas between polygons or hexagons indicate locations without mapped buildings (or water bodies); they are outside the building-based index rather than “zero risk.”

Instead, it complements these approaches by identifying where observed landscape and built-environment characteristics make flood impacts more likely, thereby supporting screening, prioritisation, and risk communication.

By maintaining consistent cell sizes, the H3 grid enables more accurate intra-urban comparisons. This contrast is visually explicit in Fig. 6, where a single sub-watershed polygon (Fig. 6a) assigns a uniform risk status to a broad area. By superimposing the H3 grid (Fig. 6b/c), we reveal that this 'uniform' catchment contains distinct clusters of high-susceptibility depressions alongside elevated safe zones. The hexagonal discretization thus recovers the spatial heterogeneity that traditional hydrological catchments smooth over.

Prior work has shown that square-grid representations can introduce corner-adjacency artifacts and visually abrupt boundaries, whereas hexagonal grids provide a more uniform neighborhood structure and smoother spatial transitions for mapping applications [19,24,25,32]. In contrast, hexagons reduce artificial segmentation and provide smoother transitions, making them more effective for identifying spatial gradients of flood risk [20,26]. Unlike square grids, hexagons allow each cell to share borders equally with all its neighbors, eliminating the corner-adjacency bias that can skew risk estimation in raster-based methods.

Catchment-based hydrologic modelling remains essential for converting precipitation (and snowmelt) into runoff volumes and discharge estimates, and is therefore central to operational forecasting at all spatial scales. In this context, the susceptibility framework presented here can be interpreted as an impact likelihood layer, indicating where generated runoff is most likely to cause damages based on learned relationships between historical impacts, topography, and built-environment characteristics. A practical hybrid approach is therefore to combine hydrologic forecasts with data-driven susceptibility mapping, and, where available, hydraulic modelling of routing and water depths, to strengthen early-warning and decision-support systems.

4.4. Visualization standards, credibility checks, and interoperability

The susceptibility maps are visualized using a consistent color palette and fixed class thresholds to ensure that PFI_h values are interpreted consistently across spatial resolutions and zoom levels. Using the same visual standard across maps reduces cognitive load and supports comparison between local hotspots and city-scale patterns, which is important for operational communication across technical and non-technical stakeholders. The class thresholds follow the scheme established in prior Norwegian applications where insurance-claim information informed interpretation and operational screening practices [26].

Although this paper does not present a full claim-based validation for the New York demonstration area, hexagon-level aggregation supports transparent credibility checks and operational plausibility assessment. Aggregating building-level predictions into PFI_h reduces sensitivity to isolated input noise and makes spatial patterns easier to compare with external evidence such as historically damaged locations, municipal incident logs, and practitioner experience. Where such reference information is available, repeated alignment between high PFI_h zones and observed flood impacts supports confidence in the susceptibility product, while systematic divergence can indicate missing predictors, data quality issues, or the need for local adaptation [43,44].

Finally, because H3 provides a common spatial index, it can function as an interoperability layer for combining susceptibility with exposure, infrastructure, and socioeconomic datasets, thereby supporting multi-criteria prioritisation and decision workflows across agencies and sectors.

4.5. Implications for disaster risk reduction practice

The operational value of PFI_h depends on the decision context. Fine resolutions support hotspot detection but are more sensitive to noise and within-cell heterogeneity, while coarser resolutions improve stability and communication but may mask critical micro-scale risks. For safety-critical screening (e.g., routing, critical assets), conservative summaries such as P95/max and fine-resolution views are recommended, whereas strategic planning products may prioritize mean-based aggregation for robustness [52]. The multi-resolution nature of the H3 grid enables the same underlying flood-susceptibility information to be communicated and analyzed at decision-relevant spatial scales without changing the spatial framework. At fine scales, they can support preparedness activities such as identifying local "susceptibility hotspots" and prioritizing property-level or block-level measures; at neighborhood and city scales, they can support drainage upgrade targeting and land-use planning; and at coarser scales, they can support strategic screening and investment prioritisation. In line with literature, susceptibility maps become most valuable when they are translated into practical tools that support planning, risk communication, and routing/location decisions (e.g., web susceptibility maps and route-planning tools) [17,53,54].

H3 cells can be aggregated and disaggregated while preserving a consistent spatial index; the resulting PFI_h products can be interpreted at multiple decision-relevant scales. At finer resolutions, the maps support identification of localized clusters that may inform property- and block-level screening and targeted interventions. At intermediate resolutions, the maps enable neighborhood-scale prioritisation that can support municipal mitigation planning, such as drainage upgrade targeting and land-use planning. At coarser resolutions, the maps provide generalized vulnerability patterns that may support regional screening and strategic allocation of mitigation resources [55].

A critical application of the PFI_h index is the protection of vulnerable institutions, such as schools and healthcare facilities, where the ability to identify *safe locations* and *safe access routes* can be decisive during pluvial flood events. For example, the street-scale resolution shown in Fig. 4d is fine enough to distinguish which side of a road is inundated or which specific intersections remain passable. This level of granularity is essential for the 'micro-scale safety planning' advocated by Youssef et al. [23], as it allows emergency planners to route evacuation corridors through specific low PFI_h cells, avoiding the 'blocked' high-risk segments that coarser maps would obscure.

Traditional catchment-scale flood maps may be appropriate for watershed-level screening, but they often lack the spatial resolution

required to distinguish safe corridors within a neighborhood. By using H3 grids at Level 11 or 12, the PFI_h system can support classification of short street segments and local areas as relatively “safe” or “unsafe” under pluvial flood susceptibility, enabling targeted evacuation planning and local response routing. This directly addresses the safety gap highlighted by Karpouza et al. [17], who emphasized that determining specific safe locations and routes is essential for protecting students during flood events. In addition, the H3 framework enables emergency planners to query within-hexagon extremes (e.g., a “maximum risk” statistic) to ensure that a designated safe zone is not compromised by a localized risk such as a single deep depression, an issue that can be obscured when using mean-based aggregation alone.

4.6. Implications of resolution and aggregation choice

The resolution sensitivity has direct implications for the interpretability and operational use of PFI_h maps (Fig. 5). First, hotspot detection degrades under coarsening: fewer cells cross high-risk thresholds, increasing the likelihood of false negatives and masking micro-clusters that can matter for maintenance, permitting, and targeted mitigation. This degradation is not merely theoretical; our spatial persistence analysis (Fig. 5b) confirms that only 14% of the high-susceptibility hotspots identified at the street scale (R13) remain visible when the map is zoomed out to the neighborhood scale (R10). This low Jaccard overlap ($J = 0.14$) warns that relying on intermediate resolutions for safety-critical screening may yield significant false negatives. Second, prioritisation can change because the distribution tails influence rankings; coarsening can reorder which neighborhoods or corridors appear most vulnerable. Third, equity and cost signals may blur, since small pockets of exposure, often where consequences are concentrated, can be under-represented when averaged into large cells.

PFI_h is computed as the mean of building-level PFI_b values within each H3 cell. As with any spatial aggregation, mean-based summarisation reduces within-cell variance and may attenuate extremes; localized susceptibility hotspots can be masked when high- and low-risk buildings co-occur within the same hexagon. As demonstrated quantitatively in Fig. 5a, this aggregation process mechanically compresses the probability distribution. While the street-level grid (R13) captures the full bimodal nature of risk, distinguishing clearly between safe and dangerous buildings, the municipal-level grid (R6) pulls all values toward the mean, effectively ‘hiding’ the statistical tails where the most severe risks reside. This behavior is consistent with the broader literature on aggregation effects and the modifiable areal unit problem (MAUP), where analytical results can change with the choice of spatial units and resolution [42,44,56,57]. Spatial resolution is also known to be particularly influential for localized flood predictions (e.g., water depth), reinforcing the need for careful interpretation when coarsening spatial representations [58]. Operationally, the smoothing effect can be mitigated by using finer H3 resolutions for hotspot-sensitive tasks and by complementing the mean with within-cell “extreme-preserving” summaries such as the 95th percentile (P95), maximum, or the share of buildings above a high-risk threshold, statistics that are standard in zonal-aggregation workflows.

From a practical implementation perspective, a useful workflow is to compute and store indicators at a high resolution for traceability, while selecting coarser resolutions only where stability and communication are the primary objectives. The appropriate resolution and aggregation statistics should therefore be chosen to match the decision context (e.g., conservative screening of critical assets may favor P95 or maximum values, whereas strategic dashboards may favor the mean for stability).

4.7. Limitations and future considerations

Central limitation of this study is that the building-level susceptibility model (PFI_b) was trained using Norwegian insurance-claim data and associated geospatial predictors within the InzureFlood project and is applied to the New York demonstration area without local retraining. Differences in building typologies, drainage infrastructure, reporting practices, and data quality may therefore influence model behaviour when transferred to new contexts. In addition, insurance claims reflect exposure, insurance coverage, and reporting practices, and do not represent complete flood occurrence. Results should consequently be interpreted as relative susceptibility patterns intended for screening and prioritisation, rather than as estimates of hazard frequency, event probability, or expected losses.

Despite its practical advantages, the H3-based workflow also introduces several methodological and operational constraints. Data availability and quality remain critical: reliable susceptibility outputs depend on consistent damage reporting and high-resolution geospatial datasets, and gaps in coverage may reduce precision or comparability across areas. Dynamic risk environments further challenge static susceptibility layers, as urban development, infrastructure upgrades, and changing precipitation regimes can shift pluvial flood susceptibility over time; periodic updates and recalibration are therefore required to maintain relevance. Comparable DGGS/H3-based workflows have been demonstrated for scalable flood detection and updating from Earth-observation time series, illustrating the feasibility of continuously updated grid-based products [59]. Finally, resolution trade-offs must be handled explicitly: finer hexagons increase local specificity but can raise privacy concerns in high-resolution public-facing products, whereas coarser hexagons improve anonymity and interpretability but may mask localized hotspots and dilute extremes.

Future development should focus on integrating susceptibility mapping with dynamic forcing and operational monitoring. Early-warning implementations would benefit from coupling static susceptibility layers with precipitation nowcasting and rainfall-runoff (hydrologic) modelling to translate meteorological forcing into time-varying runoff conditions before impacts occur. Hybrid approaches that combine catchment-based runoff generation with fine-scale vulnerability representation can offer added fidelity, particularly in dense urban environments [39]. Further integration with real-time rainfall sensors, crowdsourced incident reports, and digital twins represents a promising pathway for enhancing situational awareness and decision support. This aligns with the hybrid approach advocated by Youssef et al. [23], where static susceptibility information is combined with dynamic real-time data to support

operational early warning and prioritisation.

4.8. Summary of findings in relation to the research questions

This study investigated whether building-level scores (PFI_b) produced by the machine-learning model can be aggregated within H3 cells to yield a scalable hexagon-level index (PFI_h). The results show that this transformation is feasible: PFI_b values can be consistently assigned to H3 cell IDs and aggregated to produce interpretable PFI_h maps across multiple resolutions. Regarding Spatial Suitability, H3-based representation captures meaningful spatial clustering at fine resolutions, while coarser aggregation increasingly smooths extremes and substantially reduces hotspot persistence (e.g., Jaccard overlap of 0.14 between hotspots at R13 and R10). For Analytical Usability, aggregating PFI_b into PFI_h preserves key susceptibility information while improving interpretability for diverse stakeholders by supporting consistent visualization from local screening to city-wide overviews. Finally, for Operational Feasibility, H3-based visualizations support scalable spatial querying and updating and provide a common spatial index that enables dynamic integration of new information (e.g., flood claims or land-use changes) into decision-support workflows.

5. Conclusions

This study investigated whether building-level pluvial flood susceptibility scores (PFI_b), derived from a machine-learning approach, can be aggregated into a multi-resolution hexagon-level index (PFI_h) using the H3 discrete global grid system. The results demonstrate that this transformation is feasible and provides a scalable mechanism for communicating fine-scale susceptibility consistently across map scales [18,20]. Hexagonal aggregation also offers practical advantages for spatial analysis, including reduced edge effects and a more uniform neighborhood structure than square-based tessellations, which supports coherent interpretation of spatial patterns across resolutions [21,32]. By leveraging high-resolution geospatial predictors rather than extensive hydrodynamic parameterization, the framework illustrates a shift toward scalable, data-centric approaches to flood risk assessment [41] and aligns with policy directions emphasizing adaptable, data-informed risk management [60].

Beyond visualization, H3 indexing provides an efficient spatial reference for integrating susceptibility with complementary datasets (e.g., exposure, infrastructure, and population indicators), supporting multi-criteria prioritisation and iterative updates as new information becomes available. Where independent reference information was available, higher aggregated susceptibility aligned with historically observed flood impacts, supporting the credibility of the susceptibility product for operational screening [26]. Finally, identifying areas with high predicted susceptibility but limited recorded damage supports anticipatory planning consistent with adaptive flood management principles [6]. Overall, ML-derived building susceptibility combined with H3 multi-resolution mapping offers a flexible decision-support framework for screening, communication, and prioritisation of pluvial flood risk across stakeholder needs and spatial scales.

CRediT authorship contribution statement

Werner Svellingen: Writing – original draft, Visualization, Formal analysis, Conceptualization. **Geir Torgersen:** Writing – review & editing. **Oddbjørn Bruland:** Writing – review & editing. **Tone Muthanna:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Werner Svellingen reports financial support and article publishing charges were provided by The Research Council of Norway. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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